LINE INTEGRALS

From mathematical paths to AI decisions

Week 7 | Multivariable Calculus
Learn • Solve • Extend • Improve

Learning outcomes

START HERE

- Explain a line integral as accumulated quantity along a path.
- Distinguish scalar line integrals $\int_C f\,ds$ from vector line integrals $\int_C \mathbf{F}\cdot d\mathbf{r}$.
- Parameterize a curve and convert a line integral into an ordinary integral.
- Interpret the result in AI contexts such as robot navigation, path cost and model optimization.
- Use Python to approximate a line integral and compare paths.

Big idea

A line integral adds many small contributions made along a route.

AI connection: a robot must choose a route

MOTIVATE

Route A

Short route, but it crosses high-cost terrain.

Route B

Longer route, but the terrain is easier to cross.

AI decision

Estimate the total cost along each route; choose the route with lower cost.

Question: Is the shortest geometric path always the lowest-cost path?

A line integral provides a mathematical way to accumulate cost, energy or work along a continuous path.

Our learning method: Learn • Solve • Extend • Improve

METHODOLOGY

LEARN

Understand the meaning, notation and real-world model.

Concept

SOLVE

Parameterize the curve and evaluate the integral step by step.

Calculation

EXTEND

Change the path, field or assumptions; predict what changes.

Investigation

IMPROVE

Use computation to compare routes and justify a better decision.

AI application

Applications

Line integral means adding something along a path.

Situation	What is added?
Road trip	Fuel cost
Robot path	Energy
AI optimization	Loss change
Physics	Work

What is a line integral?

LEARN

A line integral accumulates a quantity along a curve C.

Scalar line integral

$\int_C f(x,y,z)\, ds$

Adds a scalar quantity per unit length: e.g., sensor intensity, heat exposure or route cost.

Vector line integral

$\int_C \mathbf{F} \cdot d\mathbf{r}$

Measures the component of a vector field along the direction of motion; e.g., work by a force.

Remember: ds is a small length; dr records the direction and displacement.

Scalar line integral: accumulating route risk

Suppose a robot's local risk score is f(x,y) = x + y + 1.

- At each point, f(x,y) describes risk per unit distance.
- ds is a tiny piece of the route length.
- $\int_C f(x,y)ds$ adds risk contributions along the whole route.

AI interpretation

The route with the smallest $\int_C f(x,y)ds$ may be preferable even if it is not the shortest route.

Vector line integral: work along a path

For a vector field F = P(x,y)i + Q(x,y)j:

$$\int_c f(x,y).dr = \int_c P(x,y)dx + \int_c Q(x,y)dy$$

- Positive contribution: the field helps motion.
- Negative contribution: the field opposes motion.
- The dot product F · dr measures the field component along the movement.

Applications: robot motion, wind resistance, mechanical work and force-aware planning.

Line Integral

Let **C** be a smooth space curve with initial point **A** and terminal point **B**, parameterized by its arc length **s**:

$$r(s) = x(s)i + y(s)j + z(s)k, \qquad a \le s \le b$$

(so that $r(a) = A$ and $r(b) = B$)

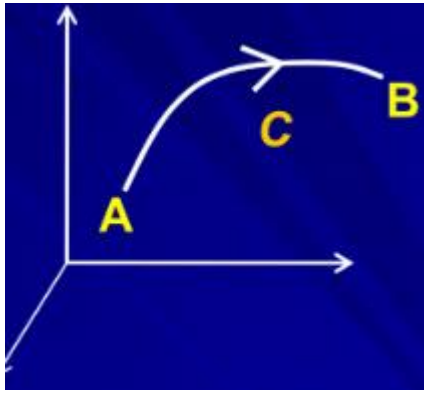
And let $f(x, y, z)$ be a scalar field defined at each point on **C**.

The **line (curve) integral** of f along **C** from **A** to **B**, with respect to arc length s, is defined by:

$$\int_C f(x, y, z)ds = \int_a^b f\big(x(s), y(s), z(s)\big)ds$$

If **C** is closed, the line integral is denoted by:

$$\oint f(x, y, z)ds$$



Method to solve Line Integral

SOLVE

1. Identify the vector field F = (P,Q) and the path C.
2. Parameterize the path: x = x(t), y = y(t), with correct limits.
3. Compute dx/dt and dy/dt.
4. Substitute into P dx + Q dy, then integrate over the parameter interval.
5. Interpret the sign and units in the context of the problem.

Worked example: a simple path for an AI agent

SOLVE

Let $F(x,y) = (x + y)i + (x - y)j$.

Find $\int_C F \cdot dr$ along the straight line from (0,0) to (1,1).

Path

$y = x$, $0 \le x \le 1$
So $dy = dx$.

Line-integral form

$F \cdot dr = (x+y)dx + (x-y)dy$

Try first: substitute $y = x$ and simplify before integrating.

Worked example: solution

SOLVE

- $F \cdot dr = (x + y)dx + (x - y)dy$
- Along $y = x$, $dy = dx$.
- Substitute: $(x + x)dx + (x - x)dx = 2x\,dx$.
- Therefore, $\int_C F \cdot dr = \int_0^1 2x\,dx = [x^2]_0^1 = 1$.

Interpretation

The line integral equals 1 in the chosen model's work/cost units.

Examples

A robot is moving in a 2D environment where a **force field** is defined as:

$$F(x, y) = (x + y)i + (x - y)j$$

This force field represents:

- Environmental resistance
- Energy consumption pattern
- External influence (like wind or terrain)

Find the **total work done (energy used)** by the robot when it moves along:

Part A) A curved path (parabola): $y = x^2$ from point **(1,1)** to **(4,2)**

Mathematical Model

Work done is given by the **line integral**:

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

Interpretation for Students

At every small step:

- Force acts on robot
- Robot moves slightly
- Small work is done

Line integral adds all these small contributions

Solution

As

$$F(x,y) = (x+y)i + (x-y)j$$
$$r = xi + yj$$
$$dr = dxi + dyj$$

We know:

$$\mathbf{F} \cdot d\mathbf{r} = (x+y)dx + (x-y)dy$$

Since path is:

$$y = x^2 \Rightarrow dy = 2x\,dx$$

Substitute:

$$= (x + x^2)dx + (x - x^2)(2x\,dx)$$

$$= (x + 3x^2 - 2x^3)\,dx$$

Solution

$$\int_1^4 (x + 3x^2 - 2x^3)\, dx$$

$$= \left[\frac{x^2}{2} + x^3 - \frac{x^4}{2}\right]_1^4$$

At $x = 4$:

$$\frac{16}{2} + 64 - \frac{256}{2} = 8 + 64 - 128 = -56$$

At $x = 1$:

$$\frac{1}{2} + 1 - \frac{1}{2} = 1$$

$$\int_1^4 f dx = -56 - 1 = -57$$

Total Work Done=-57

This value represents **total energy consumed by the robot**
It depends on Path taken or Field behavior

Python Code

```python
import numpy as np

# Parameter t = x from 1 to 4
t = np.linspace(1, 4, 1000)

# Parametrization
x = t
y = t**2

# Derivatives
dx = np.gradient(x, t)
dy = np.gradient(y, t)

# Vector field components
F1 = x + y        # (x + y)
F2 = x - y        # (x - y)

# Line integral: F.dr = F1*dx + F2*dy
integrand = F1 * dx + F2 * dy

# Numerical integration
integral = np.sum(integrand * (t[1] - t[0]))

print("Approximate value of line integral:", integral)
```

```
Approximate value of line integral: -57.11105700294975
```

Extend

What if the robot takes a different path?
Part (b): Circle $x^2+y^2=1$

Will work be same or different?

Why?

Think

1. Why is the work dependent on path here?
2. Can we reduce energy by choosing a better path?
3. How is this related to AI path optimization?

Sol of part b

$$\mathbf{F} = (x + y)\mathbf{i} + (x - y)\mathbf{j}$$

Curve:

$$C : x^2 + y^2 = 1 \quad \text{(unit circle)}$$

We evaluate:

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

Parametrize the Circle

$$x = \cos t, \quad y = \sin t, \quad 0 \le t \le 2\pi$$

$$dx = -\sin t\, dt, \quad dy = \cos t\, dt$$

$$\mathbf{F} \cdot d\mathbf{r} = (x + y)\, dx + (x - y)\, dy$$

Substitute:

$$= (\cos t + \sin t)(-\sin t\, dt) + (\cos t - \sin t)(\cos t\, dt)$$

Sol of part b

Simplifv

First term:

$$-(\cos t + \sin t)\sin t = -\cos t \sin t - \sin^2 t$$

Second term:

$$(\cos t - \sin t)\cos t = \cos^2 t - \sin t \cos t$$

Add:

$$= \cos^2 t - \sin^2 t - 2\sin t \cos t$$

Integrate

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (\cos^2 t - \sin^2 t - 2\sin t \cos t)\, dt$$

Use identities:

$$\cos^2 t - \sin^2 t = \cos 2t, \quad 2\sin t \cos t = \sin 2t$$

So:

$$= \int_0^{2\pi} (\cos 2t - \sin 2t)\, dt$$

Sol of part b

As

$$\int_0^{2\pi} \cos 2t \, dt = 0, \quad \int_0^{2\pi} \sin 2t \, dt = 0$$

Answer is 0.

Conclusion:
Same field, different path → different result.
That’s why path selection is important in AI optimization.

Your Task

Write code for **circle case (part b)**

Line Integral: Conservative vs Non-Conservative

$$\textit{Line Integral: } \int_C F \cdot dr = \int_C P\,dx + Q\,dy$$

Conservative Field

- Integral depends only on the endpoints A and B
- Path independent: $\int_C F \cdot dr = \varphi(B) - \varphi(A)$
- Closed loop: $\oint F \cdot dr = 0$
- Test: $\partial Q/\partial x = \partial P/\partial y$

Non-Conservative Field

- Integral depends on the path taken
- Different paths give different values
- Closed loop: $\oint F \cdot dr \neq 0$
- Test: $\partial Q/\partial x \neq \partial P/\partial y$

Does the answer depend on the path?

EXTEND

Scalar integral ∫C f ds

Usually depends on the route because route length and sampled values can change.

Vector integral ∫C F · dr

May depend on the path. For a conservative field, it depends only on endpoints.

Important: do not assume every vector field is conservative.

For F = (P,Q) on a suitable simply connected region, check whether ∂P/∂y = ∂Q/∂x.

AI mini-investigation: compare two routes

EXTEND

Use F(x,y) = (x + y, x − y) and travel from (0,0) to (1,1).

Route A: diagonal

$y = x,\ 0 \le x \le 1$
Compute $\int_C F \cdot dr$.

Route B: two segments

$(0,0) \to (1,0) \to (1,1)$
Compute each segment, then add.

Prediction: both results are equal because this field is conservative.

Verify the prediction by calculation; do not rely on intuition alone.

Why the two routes give the same work

EXTEND

- $P(x,y) = x + y$, $Q(x,y) = x - y$
- $\partial P/\partial y = 1$ and $\partial Q/\partial x = 1$.
- The partial derivatives match; on $\mathbb{R}^2$ this field is conservative.
- A potential function is $\phi(x,y) = x^2/2 + xy - y^2/2$.

Fundamental theorem for line integrals

$\int_C F \cdot dr = \phi(1,1) - \phi(0,0) = 1$.

Where line integrals fit in an AI workflow

IMPROVE

SENSE

Collect terrain, obstacle or sensor data.

→

MODEL

Represent local cost or force as a field.

→

EVALUATE

Integrate cost/work along candidate routes.

→

OPTIMIZE

Compare routes and select a suitable one.

The integral evaluates a candidate path; an optimization algorithm searches for a better path.

Python code: approximate a vector line integral

IMPROVE

```
import numpy as np

# Route r(t) = (t, t), 0 <= t <= 1
t = np.linspace(0, 1, 10001)
x, y = t, t

dx_dt = np.ones_like(t)
dy_dt = np.ones_like(t)

P = x + y
Q = x - y
integrand = P * dx_dt + Q * dy_dt

value = np.trapz(integrand, t)
print(value)   # approximately 1.0
```

Experiment

Increase the number of sample points.

Compare

Check numerical output against the exact answer.

Debugging challenge: check the model before coding

A path is stated as $y = x^2$ from (1,1) to (4,2). Is this consistent?

- At x = 4, y = 16, not 2.
- Therefore, the stated endpoint (4,2) is not on the curve.
- Correct the endpoint or choose a path that connects the stated points before integrating.

AI practice

In Artifical Intelligence, validate the inputs and assumptions before trusting the output.

Guided practice: calculate, then interpret

SOLVE

Let F(x,y) = (2x, y) and C be the straight line from (0,0) to (1,2).

- Parameterize C using x = t, y = 2t, 0 ≤ t ≤ 1.
- Find dx and dy in terms of dt.
- Evaluate ∫C F · dr.
- Explain what the sign of your result means in a work interpretation.

Hint: ∫C F · dr = ∫ (2x dx + y dy).

Methodology checkpoint

LEARN

Explain what the integral accumulates.

SOLVE

Set up and evaluate the integral.

EXTEND

Change the path and test your prediction.

IMPROVE

Use code and evidence to make a decision.

Example

A software engineering team is developing a network monitoring system to estimate the cost of transmitting data between two servers. The transmission cost depends on the network state, represented by two variables, (x) and (y), where (x) represents the normalized data load and (y) represents the normalized network congestion.

The team models the transmission cost using the vector field

$$A = (1 + x^2y)i + 2xyj$$

where the two components represent the cost contributions associated with changes in data load and network congestion.

The team wants to calculate the total transmission cost when data moves from Server A, located at ((0,0)), to Server B, located at ((1,1)). The data may travel through different network routes.

The total cost along a route (C) is determined by the line integral

As a software engineering student, you are required to analyze the transmission cost for the following three network routes.

Route 1: Direct transmission

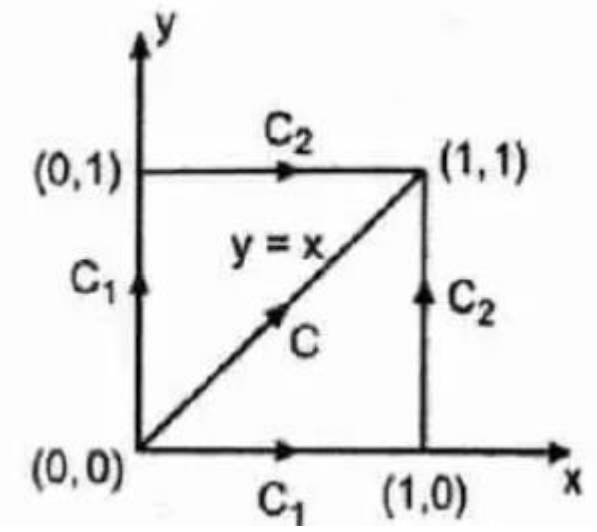
Data travels directly from Server A ((0,0)) to Server B ((1,1)) along the straight-line path (y=x).

Route 2: Horizontal and vertical transmission

Data first travels horizontally from ((0,0)) to ((1,0)), then vertically from ((1,0)) to ((1,1)).

Route 3: Vertical and horizontal transmission

Data first travels vertically from ((0,0)) to ((0,1)), then horizontally from ((0,1)) to ((1,1)).



Example

Tasks

1. Express the line integral in terms of (dx) and (dy).
2. Calculate the total transmission cost along Route 1.
3. Calculate the total transmission cost along Route 2 by evaluating both line segments separately.
4. Calculate the total transmission cost along Route 3 by evaluating both line segments separately.
5. Compare the results for all three routes and explain whether the total transmission cost depends on the selected route.
6. Develop a Python program that calculates and verifies the transmission cost for all three routes using symbolic or numerical integration.

Case Study

An AI system is evolving in a learning environment where the training dynamics are represented by the vector field:

$$A = 2x^2yi + 2(x^2y + y)j$$

Evaluate the **total learning effort (work done)** required to move the AI model from the initial state **(0,0)** to the target state **(2,4)** under the following training strategies:

1.Linear Growth Strategy:

The model improves proportionally, following the relationship:

$$y = 2x$$

2. **Accelerated Learning Strategy:**

The model follows a nonlinear improvement path:

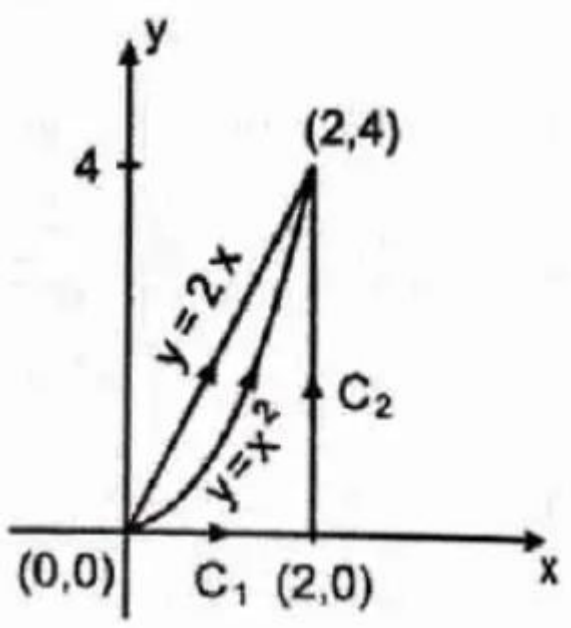
$$y = x^2$$

3. **Stepwise Deployment Strategy:**

First, scale horizontally from **(0,0) to (2,0),**

Then, scale vertically from **(2,0) to (2,4)**

Task: Determine the **total learning effort** for each strategy and assess whether the effort depends on the chosen training path.



Case study

A company operates in a dynamic environment where **performance drivers** (cost pressures, efficiency, and revenue actions) are modeled by:

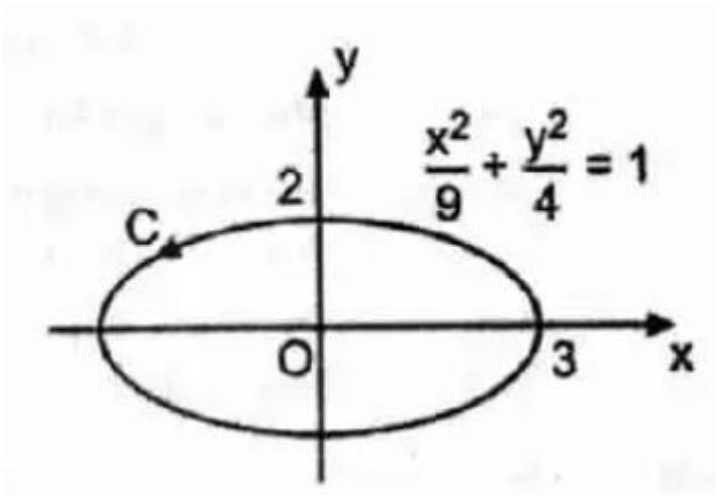
$$A = (x - 3y)i + (y - 2x)j$$

The organization completes a **full annual performance cycle** consisting of:

- Budgeting
- Execution
- Monitoring
- Review & corrective actions

This full cycle forms a **closed loop C** of operational performance, represented by:

$$\frac{x^2}{9}+\frac{y^2}{4} = 1$$



The cycle is executed in a **forward-moving (counterclockwise) direction**, reflecting continuous improvement.

Evaluate the **net organizational impact (value created or lost)** after completing one full business cycle.

Solution

Given

Vector field:

$$A = (x - 3y)\,\hat{i} + (y - 2x)\,\hat{j}$$

Closed curve C:

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \quad \text{(ellipse, counterclockwise)}$$

We need to evaluate:

$$\oint_C A \cdot d\mathbf{r}$$

solution

$$A = (x - 3y)i + (y - 2x)j$$

As $$r = xi + yj$$

$$dr = dxi + dyj$$

$$A.dr = (x - 3y)dx + (y - 2x)dy$$

The parametric equations of this ellipse

$$x = 3cost, \quad y = 2sint, \quad 0 \leq t \leq 2\pi$$

$$dx = -3sint\ dt, \qquad dy = 2cost\ dt$$

$$\oint_C \bar{A} \cdot d\bar{r} = \int_{t=0}^{2\pi} (3\cos t - 6\sin t)(-3\sin t\, dt) + (2\sin t - 6\cos t)(2\cos t\, dt)$$

$$= \int_0^{2\pi} (-5\sin t\cos t + 18\sin^2 t - 12\cos^2 t)\, dt$$

$$= \int_0^{2\pi} \left[-\frac{5}{2}\sin 2t + 9(1-\cos 2t) - 6(1+\cos 2t) \right] dt$$

$$= \left| \frac{5}{4}\cos 2t + 9\left(t - \frac{\sin 2t}{2}\right) - 6\left(t + \frac{\sin 2t}{2}\right) \right|_0^{2\pi}$$

$$= \left| \frac{5}{4}\cos 2t + 3t - \frac{15}{2}\sin 2t \right|_0^{2\pi}$$

$$= \left(\frac{5}{4}\cos 4\pi + 6\pi\right) - \left(\frac{5}{4}\cos 0\right)$$

$$= \frac{5}{4} + 6\pi - \frac{5}{4} = 6\pi$$

Result

The value is **positive (6π)**
That means the system is **generating net value over the cycle**